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10 October 2026

Dr Daniel Lee
Department of Chemical Engineering, School of Engineering
The University of Manchester
Manchester M13 9PL, United Kingdom

Re: Research Associate in Solid-State NMR Spectroscopy, ref. SAE-032399

Dear Dr Lee,

I am writing to apply for the position of Research Associate in Solid-State NMR Spectroscopy (ref. SAE-032399) within the EPSRC Programme Grant on Molecular Solid Solutions. I completed my PhD in Chemical Sciences at the Institute of Chemistry, UNAM (Mexico), in 2026 under the supervision of Prof. Braulio Rodríguez-Molina, and the degree was awarded with honours.

Solid-state NMR was the central technique of my PhD. On a Bruker 500 MHz spectrometer with a double-resonance (^{1}H/X) MAS probe, I set up my own experiments, including tuning and matching both channels and retuning the ^{1}H channel for ^{19}F. I acquired ^{13}C CP-MAS spectra with variable contact times, ^{19}F MAS spectra, variable-temperature ^{2}H spectra and some ^{15}N CP-MAS spectra. I simulated ^{2}H lineshapes with NMR WebLab to obtain rotational rates and activation energies, and analysed ^{13}C T_1 and ^{1}H and ^{19}F relaxometry data to follow faster motions.

I use NMR together with diffraction. In a pyrrolo[3,2-b]pyrrole solvate that jumps on heating (Mater. Chem. Front., 2026), ^{2}H NMR and variable-temperature single-crystal diffraction allowed me to separate the molecular dynamics from the anisotropic thermal expansion that drives the jump. In a zirconium MOF (ACS Mater. Lett., 2026), the same combination showed how a photoswitch and guest uptake change the motion of a molecular rotor. Both projects were done with Dr Martin Dračínský and Dr Jan Blahut at IOCB Prague.

The role also involves preparing the solid solutions, and synthesis and crystallisation are a large part of my training. I grow single crystals of molecular crystals, cocrystals and solvates and characterise them by SCXRD, PXRD, DSC, TGA and hot-stage microscopy. I contributed to a study in which the salient temperature of cocrystals was tuned through the composition of the crystallisation solvent mixture (Angew. Chem. Int. Ed., 2026). I also run DFT (Gaussian16, periodic CP2K) and molecular dynamics (LAMMPS) on HPC clusters. Diffraction gives the average structure of a molecular solid solution, while solid-state NMR can show how a dopant is distributed and how it moves locally, and this is the question I would like to work on within MoSS.

I have not yet used DNP. The UK DNP MAS NMR Facility at Manchester, and the chance to extend my work to ^{17}O and natural-abundance ^{15}N, are among my main reasons for applying. During my PhD I learned periodic DFT and molecular dynamics from scratch and now use both routinely, and I would approach DNP in the same way.

My 14 publications include collaborations with groups in the Czech Republic, the United States, China and the United Arab Emirates. I co-supervised four bachelor's theses and nine undergraduate research placements, trained students in laboratory procedures and safety, and handled purchasing and inventory for our laboratory. I have presented 15 posters and 9 talks, and received the Best Poster Award at DynaMIC23 in Fribourg. Thank you for considering my application; I would be glad to discuss it at interview.

Yours sincerely,

Dazaet Galicia-Badillo, PhD